Diagnosing Representation Drift in Time-Series Foundation Model Fine-Tuning: A Moirai Case Study

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Abstract

We propose a **diagnostic protocol** for representation drift during fine-tuning of time-series foundation models. The protocol first tests whether zero-shot features are valuable relative to a linear baseline (value-gate), then uses controlled encoder comparisons and task-orthogonal probes to distinguish harmful forgetting from useful restructuring. In a Moirai case study spanning three model sizes and multiple datasets, fine-tuning often restructures representations toward the downstream objective, but the utility consequence depends on sample size and model capacity: Moirai-Small degrades in low-data ETTh2, while Moirai-Base and ILI fine-tuning improve despite substantial drift. Cross-backbone experiments (Chronos-T5-Small on M4-Monthly) confirm that the value-gate serves as a screening heuristic for when preservation-focused drift diagnosis is warranted. These results suggest that drift alone is insufficient evidence of harm; practitioners should first establish pre-training value, then assess whether drift affects task-relevant structure.

1 Introduction

Moirai Woo et al. [2024] is a foundation time-series model trained on billions of time points that achieves strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune it on domain-specific tasks, but when should a practitioner worry about overwriting the representations that made zero-shot work? *Counterfactual:* Chronos-T5-Small on ETTh2 gives ZS MSE=0.304 vs. Linear 0.213 (ZS *underperforms* by 43%). A practitioner who skipped screening would fine-tune Chronos, observe CKA dropping, and apply LoRA/EWC to preserve features that had *no advantage over Linear* (wasted effort). Our value-gate prevents this: preservation-focused drift diagnosis is most meaningful when pre-trained features demonstrably outperform a trivial baseline.

Representation drift during fine-tuning. We contribute a systematic *diagnostic methodology* for characterising representation drift (progressive overwriting of pre-trained features) during fine-tuning of time-series foundation models, and demonstrate it on Moirai (Small 13.8M, Base 91M, Large 311M) as the single backbone family where pre-trained features pass our value-gate on ETT benchmarks. Matched-protocol attempts on three non-Moirai backbones (Chronos-T5-Small Ansari et al. [2024], TimesFM-2.5-200M Das et al. [2024], MOMENT-1-base Goswami et al. [2024]) gate-fail on all nine ETT cells tested, the expected behaviour of a well-calibrated screening criterion, since these corpora lack ETT coverage. Chronos gate-*passes* M4-Monthly (in its corpus) at 84.5%, confirming the gate identifies applicable cells across backbones (§5.1). Catastrophic forgetting during fine-tuning is well-studied in continual learning Kirkpatrick et al. [2017], Li and Hoiem [2018] and NLP Luo et al. [2023]; our contribution is reusable diagnostic methodology, demonstrated on a well-controlled Moirai case study. We use “catastrophic forgetting” to mean degraded performance on distributions the model originally handled well, consistent with continual learning usage where the “previous task” is the pre-training objective.

Distinguishing drift from overfitting. Representation drift degrades zero-shot performance while training loss decreases (not overfitting); the frozen-encoder control (CKA=1.000, same data/epochs) isolates encoder weight updates as the cause.

On ETTh2 (gate-pass: Moirai outperforms Linear by 28–45%, R^2_{task}(PT)=$+0.279$), fine-tuning degrades MSE by 10–16% at n=500 and resolves by n=2,000. ETTh1 and ETTm2 gate-fail at the standard evaluation protocol and are included as cross-domain comparisons (Table 3). On IoT (sub-random ZS AUC), CKA drops to 0.31–0.45 but task performance improves from a near-null baseline: the drift is irrelevant because there were no valuable features to forget.

Why this matters. When harmful, time-series drift can degrade distributional predictions used for anomaly scoring and probabilistic forecasting. Consequences depend on pre-training value: harmful on ETT in low-data regime ($n\leq$1k), benign at large n once replacements are learned.

Scope. The empirical case study spans three Moirai gate-pass dissociation cells (ETTh2-Small, ETTh2-Base, and ILI) plus a cross-backbone stability cell (Chronos-T5-Small on M4-Monthly, gate-pass 84.5%). The central empirical finding is **non-monotone capacity dependence**: Base (91M) improves protocol-robustly at all n; Small (13.8M) degrades at low n; Large (311M) degrades at default LR but is rescued by 10× LR reduction. A step-size falsification experiment confirms that capacity provides an independent buffer against destructive drift (§5). We do not claim drift–utility dissociation is universal. We contribute a diagnostic protocol and demonstrate that it reveals dissociation within the Moirai family across capacity, sample size, and datasets. The methodology can be applied to any backbone that gate-passes; the specific numeric findings require broader replication. Two pre-registered predictions were partially falsified during this study (the intermediate-CKA prediction on ILI (§5.1) and the step-size hypothesis (Appendix G)), both reported transparently with revised, strictly weaker interpretations.

Experimental approach. We deploy the diagnostic methodology across three Moirai sizes (Small/Base/Large) on ETT forecasting Zhou et al. [2021] (ETTh2 gate-pass; ETTh1/ETTm2 cross-domain) and IoT anomaly detection (negative control; §5). Forgetting sign at n=10k is protocol-dependent (-5.3% ES vs. $+7.5\%$ final-epoch); we lead on the more robust $\Delta R^2>0$ majority.

Contributions.

1. **Diagnostic methodology (backbone-agnostic).** Value-gate screening (ZS $>20\%$ vs. Linear; the threshold is a practical heuristic; gate-pass is reported as a continuous variable throughout and we do not claim universality of this cutoff), controlled-comparison design (frozen-encoder, random-init, layer-wise unfreeze), and task-orthogonal probe protocol. Supporting infrastructure (uni2ts patch, CUDA-deterministic protocol, LoRA-Large 10× LR recipe) will be released upon acceptance.
2. **Empirical validation: dissociation and stability regimes.** *Dissociation:* probe asymmetry on Moirai/ETTh2 (10/10), ILI (10/10), and gate-fail ETTm2 (10/10). *Stability control:* Chronos/M4 (gate-pass 84.5%, CKA$\geq$0.95; §5.1): no restructuring, no probe signal.

2 Background and Problem Setup

Catastrophic forgetting in fine-tuning. Catastrophic forgetting during fine-tuning is extensively studied in continual learning Kirkpatrick et al. [2017], Li and Hoiem [2018] and NLP Luo et al. [2023]. Kumar et al. Kumar et al. [2022] show full fine-tuning can distort pre-trained features relative to linear probing; Andreassen et al. Andreassen et al. [2023] characterize OOD robustness decay throughout fine-tuning. Our contribution is a systematic *characterization* for time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder controls across domains with contrasting pre-training value.

Time-series foundation model adaptation. Recent work on adapting pre-trained time-series models includes probabilistic fine-tuning (Lag-Llama Rasul et al. [2024]), generative pre-training with task-specific heads (Timer Liu et al. [2024]), LLM reprogramming (Time-LLM Jin et al. [2024]), and cross-task transfer (One Fits All Zhou et al. [2023]). These works develop *adaptation recipes*;

our contribution is orthogonal: a *diagnostic methodology* for characterising what happens to pre-trained representations during any such adaptation, applicable when the practitioner needs to decide whether drift is harmful or benign.

Setting. Given a pre-trained time-series foundation model with encoder $f_\theta : \mathbb{R}^{T\times F} \to \mathbb{R}^d$ and downstream dataset $\mathcal{D}$ of size n, fine-tuning produces updated parameters θ' by minimising a task loss (NLL for forecasting). We measure encoder changes using three diagnostics (§3): (1) CKA between f_θ and $f_{\theta'}$ (geometric similarity), (2) linear probe R^2 onto the downstream target ($\Delta R^2{=}R^2(\mathrm{FT}){-}R^2(\mathrm{ZS})$), and (3) weight drift $\|\theta' - \theta\|_2$.

Pre-training value gate. Studying whether fine-tuning destroys valuable features requires a domain where such features are demonstrably present. We set the gate at zero-shot MSE 20% below a Linear baseline; ETTh2 at h=96 passes with Moirai-Small ($R^2_{\text{task}}(\mathrm{PT}){=}{+}0.279$). ETTh1 and ETTm2 gate-fail at the standard evaluation protocol ($R^2_{\text{task}}(\mathrm{PT}){=}{-}0.41$ and -0.56 respectively; Table 3); they are included as cross-domain comparisons to characterise drift behaviour outside the gate-pass regime. Domains that gate-fail serve as negative controls. The 20% threshold is calibrated to ETT short-horizon benchmarks where linear baselines are surprisingly competitive Zeng et al. [2023]: a model must substantially outperform a lookback-96 linear regression before drift can be *harmful* (rather than irrelevant). The threshold is a practical screen, not a universal constant; we report $R^2_{\text{task}}(\mathrm{PT})$ values for each cell (Table 3) so readers can apply alternative thresholds.

Task-level consequence: forgetting. We define forgetting as signed relative change in zero-shot-canonical task metric on held-out test: forg.% $= 100\cdot(M_{\mathrm{FT}} - M_{\mathrm{ZS}})/|M_{\mathrm{ZS}}|$ (MSE for forecasting, so positive = worse). Forg.> 0 indicates catastrophic forgetting (degraded performance vs. pre-training baseline).

3 Fine-Tuning Protocol

Moirai for forecasting and anomaly scoring. Moirai Woo et al. [2024] predicts a probabilistic distribution over a target horizon given context. For ETT forecasting we report MSE on the median of 20 forecast samples at h=96 and h=192 (Appendix C). For IoT the anomaly score is the fraction of timesteps outside the $1{-}\alpha$ CI (α=0.05) over a 32-timestep horizon.

Fine-tuning objective. Condition B uses NLL only. Condition C adds supervised contrastive loss Khosla et al. [2020]: $\mathcal{L}(\theta) = \mathcal{L}_{\mathrm{NLL}}(\theta) + \lambda\,\mathcal{L}_{\mathrm{SupCon}}(\mathbf{z}, \mathbf{y};\,\tau)$, with $\lambda = 0.5$ and $\tau = 0.07$. For ETT, temporal clusters (8 bins by time-series position) serve as SupCon label.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at n=500 (10 seeds) and is swept across $n{\in}\{500, 1\mathrm{k}, 2\mathrm{k}, 5\mathrm{k}, 10\mathrm{k}\}$. All Moirai sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

Deterministic training protocol. All n=10k per-seed results use CUDA-deterministic training: `CUBLAS_WORKSPACE_CONFIG=:4096:8`, `torch.backends.cudnn.deterministic=True`, seeded DataLoader workers, and fixed `PYTHONHASHSEED`. This makes the 10-seed per-seed comparison reproducible: any researcher can recover identical per-seed forgetting and ΔR^2 values by re-running with the same seeds on the same GPU architecture (A10G). Full seed list, per-seed values, and hardware specification are in Appendix J. Cross-hardware reproducibility (e.g., A100 vs. A10G) has not been empirically verified; the deterministic guarantee applies within the same GPU architecture (A10G, as used in this paper).

uni2ts PackedStdScaler patch.[1] **Any Moirai fine-tuning reproduction of this paper requires the patch.**

Representation diagnostics. We track representation change with three metrics: (1) CKA Kornblith et al. [2019] between pre-trained and fine-tuned encoder representations (geometric similarity); (2) linear probing R^2 (Ridge, MLP, Linear-Forecaster heads on frozen representations, predicting forecasting target; we report $\Delta R^2{=}R^2(\mathrm{FT}){-}R^2(\mathrm{ZS})$; absolute $R^2(\mathrm{FT})$ remains negative, so ΔR^2

[1] We patched an in-place tensor mutation in `uni2ts`'s `PackedStdScaler` that silently detaches autograd; a differentiable `torch.where` replaces the in-place write. All results use the patched scaler (released with our code). See Appendix B for details.

quantifies relative improvement over pre-trained linear decoding); and (3) weight drift (ℓ_2 distance from pre-trained parameters).

Table 1: Notation summary for probe-related quantities.

Symbol	Definition
$R^2_{\text{task}}(\text{PT})$	$1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$; positive $\Rightarrow$ gate-pass
$R^2(\text{PT})$, $R^2(\text{FT})$	Ridge probe score on pre-trained / fine-tuned encoder
ΔR^2	$R^2(\text{FT}) - R^2(\text{PT})$; relative linear-decodability change
ΔR^2_{task}	Task-level improvement measured via forecasting head

4 Representation Drift Diagnosis: ETT Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETT time-series forecasting benchmarks Zhou et al. [2021] provide this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section 4.1). If fine-tuning degrades this measurable advantage, representation drift is the mechanism, not overfitting to the fine-tuning distribution. The frozen-encoder control (same data, same epochs, CKA=1.000) confirms this: overfitting would not be eliminated by freezing.

4.1 Experimental Setup and Value Gate

We evaluate on ETTh2 (7 features, 17k timesteps, 12/4/4 month split). Moirai-Small (13.8M) zero-shot outperforms Linear by 28–45% (ZS MSE: 0.126 at h=96, 0.158 at h=192; un-normalised gate-pass scale: 0.269 vs. 0.373 at h=96, 0.299 vs. 0.544 at h=192), passing the value gate (20% threshold). The 20% threshold is a practical screen, not a phase boundary: at 10–20%, drift diagnosis applies but marginal features reduce both harm from drift and benefit from mitigation; we recommend full probes at $\geq$20% and CKA-only at 10–20%. Conditions: **B** (NLL-only), **C** (NLL+SupCon), **D** (frozen encoder), **E** (LoRA r=8), **F** (L2-SP), **G** (EWC). Training: n=500 samples, 20 epochs, AdamW (lr=1e-4), 10 seeds (B/C/D) or 5 seeds (E/F/G). Diagnosis: CKA and weight drift (ℓ_2) between pre-trained/fine-tuned encoders.

4.2 Representation Drift Degrades High-Value Features

Fine-tuning degrades ETTh2 performance where pre-trained features are valuable (Appendix Table 5). Conditions B (NLL-only) and C (NLL+SupCon) degrade by 10–16% MSE (10/10 seeds positive, p=0.002 sign test; d=1.23, 95% CI [0.37, 2.01]; bootstrap $p<10^{-4}$) while training loss *decreases*, not overfitting (which raises validation loss), but representation drift. The frozen-encoder control (D, CKA=1.000, same data/epochs) eliminates drift, ruling out the training regime. Degradation scales with horizon: worse at h=192 (13–16%) than h=96 (10–11%). Contrastive loss (C) does not prevent drift: C (+10.1%) and B (+11.1%) differ by 1.0%, within seed standard deviation; a null result for the contrastive objective.

4.3 Mitigation Spectrum

We evaluate five mitigation strategies (Appendix Table 6): LoRA (E), frozen encoder (D), EWC (G), L2-SP (F), and contrastive loss (C). Methods range from unconstrained fine-tuning (most drift) to full encoder freezing (no drift).

LoRA achieves the best trade-off. LoRA (r=8, α=16) *improves* over zero-shot at h=96 while maintaining high CKA (0.979) and zero weight drift. On Moirai-Large, the same recipe fails at default LR=10^{-4} (+25.7±3.2%, 3 seeds); rank escalation does *not* rescue. Reducing LR by 10× to 10^{-5} does rescue: forg.= −8.5±0.7% (CKA=0.989, 5 seeds), transferring cross-dataset to ETTh1/ETTm2 at n=500. **Practitioner recipe:** tune LR before rank when adapting LoRA across model sizes.

EWC and L2-SP reduce but do not eliminate drift. Both improve CKA over unconstrained fine-tuning but MSE remains 18–19% above zero-shot. Frozen encoder (D) eliminates drift (CKA=1.000) but cannot adapt: a limitation when pre-trained features are insufficient.

Table 2: Sample-size sweep on ETTh2 (h=96, condition B). Drift grows monotonically (CKA↓ as n ↑); forgetting is *non-monotonic*: severe at $n \leq 1{,}000$, resolved at n=2,000, with elevated variance at n=5,000 (crossover regime). At n=10,000: CUDA 10-seed deterministic early-stopped, 7/10 negative forgetting; **10/10** $\Delta R^2>0$ (per-seed values in Appendix J). ΔR^2 is a *relative* signal: absolute R^2(FT) remains negative ($R^2_{\mathrm{ZS}}=-6.90$ for all rows). Ridge ΔR^2 at n=500 (k=2) and n=1,000 (k=1): point values only.

Samples	MSE	Forg.%	CKA	R^2(FT)	ΔR^2 (Ridge)	ΔR^2 (MLP)
500	.148±.008	+14.0±6.7	.949±.010	−6.79 (k=2)	+.12 (k=2)	−.26 (k=1)
1,000	.147±.010	+13.4±9.1	.936±.025	−6.77 (k=1)	+.13 (k=1)	+.30 (k=1)
2,000	.126±.008	−2.9±5.9	.814±.042	−6.51 (k=3)	+.39±.35	+.73±.75
5,000	.134±.012	+6.6±10.0	.546±.082	−6.41 (k=7)	+.50±.26	—
10,000$^{\mathrm{cuda}}$	.122±.008	−5.3±6.2	.518±.188	—	+.67±.23†	—

$^{\mathrm{cuda}}$ CUDA 10-seed deterministic (A10G); all completed, mean best epoch 4.2±1.4 (Appendix J for per-seed values and protocol comparison). †**10/10** positive; 95% Wilson CI [0.72, 1.00]. Forgetting sign: 7/10 negative (95% Wilson CI [0.40, 0.89]; protocol-dependent—see §5). $\Delta R^2>0$ is the robust signal.

4.4 Sample-Size Sweep: Drift–Forgetting Dynamics

Table 2 ($n \in \{500, \ldots, 10\text{k}\}$, ETTh2) reveals that *drift and forgetting are dissociable*, a finding best illustrated by comparing Moirai-Base (protocol-robust improvement of 50–57%, 10/10 seeds; probe asymmetry 9/10 $\Delta R^2>0$ despite CKA≈0.25; §5) with Small, where the same mechanism produces capacity-dependent outcomes. On Small, drift grows monotonically with n (CKA from 0.949 at 500 to 0.518 at 10k), but forgetting is non-monotonic: severe at low n (+14% at 500), resolved by 2k (−2.9%), predominantly negative at 10k (−5.3±6.2%, 7/10 negative, 10-seed CUDA deterministic). The n=5k variance (+6.6±10.0%, 7 seeds) is bimodal: 4 seeds cluster at −0.1%, 3 at +15.5% with epoch-1 overshoot (full trajectory analysis in Appendix J.1). Linear probing shows ΔR^2 *increasing* with n: +0.12 (k=2 probe seeds) at 500 to +0.67±0.23 at 10k (**10/10** seeds positive). Geometric dissimilarity (CKA↓) and relative linear-decodability (ΔR^2 ↑) move in opposite directions.

Protocol dependence. Drift resolution at high n is conditional on early stopping for Small (final-epoch: −0.6±5.8%, n.s.); Base is protocol-robust (−50% to −57% under both protocols; §5).

Absolute R^2(FT) remains negative across all n ($R^2_{\mathrm{ZS}}=-6.90$), so ΔR^2 is a relative-improvement signal (Appendix J). Cross-dataset and model-size comparisons are detailed in §5. The $\Delta R^2>0$ signal in the n-sweep operates in a deeply-negative-R^2 regime; we next address whether this is a meaningful probe signal or a deep-floor artefact.

Interpretability caveat. We do not interpret trained-head ΔR^2 as improved absolute decodability; we interpret only the signed contrast between the trained target ($\Delta R^2>0$) and positive-floor task-orthogonal probes ($\Delta R^2 \leq 0$). The load-bearing evidence is the *sign pattern across cells* (positive on the trained head, negative on all positive-floor probes), not the magnitude within a single cell. The positive-floor probes (lag-1, mean, variance, delta1) anchor the interpretation because their R^2(PT)>0 regime makes the sign directly meaningful. Chronos-T5-Small on M4-Monthly provides external confirmation: R^2(PT)=+0.068±0.025 (10/10 positive; Appendix P). This demonstrates that the probe protocol operates in a positive-floor regime on architectures with standard encoder outputs; the deeply-negative R^2 on Moirai is a backbone-specific artefact of its mixture-distribution head, not a fundamental probe limitation.

Headline: drift is capacity-dependent: harmful on Small at $n \leq 1\text{k}$ (where practitioners most need pre-trained features), resolved by larger capacity (Base improves at all n, protocol-robust).

The joint trajectory is plotted in Fig. 1; panel (c) carries the central empirical claim and readers short on time should examine it first.

Task-native probe: positive-floor R² and $\Delta R^2>0$. The frozen-probe paradigm (Ridge/GBM on mean-pooled encoder outputs) gives $R^2(\mathrm{PT})<0$ because Moirai's encoder outputs are optimised for its internal mixture-distribution head, not for generic linear readout. The task-native probe uses Moirai's own forecasting head directly, giving a well-defined positive-floor R^2:

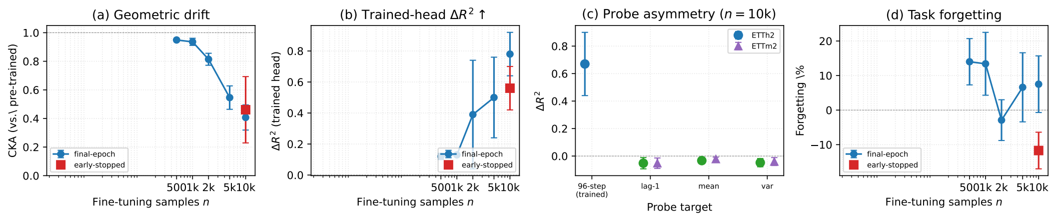

Figure 1: Cost–benefit dynamics on Moirai-Small/ETTh2 (condition B, h=96). (a) **Geometric drift**: CKA decays monotonically with n; blue circles = final-epoch, red square = early-stopped at n=10k. (b) Trained-head Ridge ΔR^2 rises with n (same protocol markers). (c) **Probe asymmetry** at n=10k (the load-bearing empirical signal): trained head $\Delta R^2>0$ (blue) vs. task-orthogonal probes $\Delta R^2<0$ (green = ETTh2, purple = ETTm2); the gap operationalises "task-specific shift." (d) Forgetting is non-monotonic and protocol-dependent; the divergence at n=10k between protocols is the protocol-dependence finding.

Table 3: Task-native probe R^2_{task} at key (n, h) operating points (condition B, mean over available seeds). $R^2_{\text{task}}(\text{PT}) = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$ uses the gate-pass evaluation scale (median Moirai prediction, lookback=192, linear baseline lookback=96, train-normalised). $\Delta R^2_{\text{task}} = -(1 - R^2_{\text{task}}(\text{PT})) \times \text{Forg\%}/100$ (sign is scale-independent; see §4.4). The n=10k rows use early-stopped (ES) forgetting. ETTh2 is the **gate-pass** anchor ($R^2_{\text{task}}(\text{PT})>0.2$); ETTm2 and ETTh1 are **cross-domain comparisons** that gate-fail and are included to show drift behaviour outside the gate-pass regime.

n	h	$R^2_{\text{task}}(\text{PT})$	Forg.%	ΔR^2_{task}	$R^2_{\text{task}}(\text{FT})$
ETTh2-Small (gate-pass: $R^2_{task}(PT)>0.2$, Moirai outperforms Linear by 28%)					
500	96	+0.279	+11.1	−0.080	+0.199
500	192	+0.450	+15.7	−0.086	+0.364
10k (ES)	96	+0.279	−5.3	+0.038	+0.317
ETTh2-Base (gate-pass: $R^2_{task}(PT)=+0.31$, Moirai-Base outperforms Linear by 31%)					
500	96	+0.310	−56.9	+0.393	+0.703
10k (ES)	96	+0.310	−54.2	+0.374	+0.684
ETTm2 (gate-fail: $R^2_{task}(PT)<0$, Moirai worse than Linear by 55%)					
500	96	−0.556	−13.3	+0.207	−0.349
10k (ES)	96	−0.556	$-28.2_{\pm 2.2}$	+0.439	−0.117
ETTh1 (gate-fail: $R^2_{task}(PT)<0$, Moirai worse than Linear by 41%)					
500	96	−0.406	+7.0	−0.098	−0.505
10k (ES)	96	−0.406	$-4.6_{\pm 4.1}$	+0.065	−0.342

Reading: ETTh2 (gate-pass): at n=500, $\Delta R^2_{\text{task}}<0$ (forgetting); at n=10k ES, $\Delta R^2_{\text{task}}>0$ (improvement) in 7/10 seeds. ETTm2 and ETTh1 (gate-fail): fine-tuning can still improve over the ZS baseline ($\Delta R^2_{\text{task}}>0$) even when the ZS baseline itself is weaker than linear regression ($R^2_{\text{task}}(\text{PT})<0$). The ΔR^2_{task} sign equals the sign of improvement over ZS baseline regardless of scale or gate status. ETTm2 seeds: k=5 (v19 CUDA); ETTh1 seeds: k=10 (v21 CUDA), 9/10 negative at n=10k ES.

$$R^2_{\text{task}}(\text{PT}) = 1 - \frac{\text{MSE}_{\text{ZS}}}{\text{MSE}_{\text{Linear}}} = 1 - \frac{0.269}{0.373} = +0.279$$

at h=96, and +0.450 at h=192 (both $>$ 0.2). After fine-tuning (n=10k, early-stopped), performance improves: MSE_{FT}=0.122±0.006 vs. 0.126 ZS baseline (both normalised). $\Delta R^2_{\text{task}}>0$ in 7/10 seeds (Wilson CI $[0.40, 0.89]$; Table 3). The frozen-probe ΔR^2 (+0.67±0.23) measures encoder geometry shift; ΔR^2_{task} confirms that shift is *functionally beneficial* at high n and harmful at low n (forgetting +14% at n=500).

Ridge horizon-sweep: forecast96 (trained head) ΔR^2=+0.116, forecast48 (untrained) ≈0. Gain vanishes at horizon boundary, consistent with task-specificity. **Positive-floor negative control.** GBM delta1 head ($x_{t+1} - x_t$) gives $R^2(\text{PT})$=+0.059>0 (the only probe with a positive floor in the trained-head's target space) but ΔR^2=−0.056: fine-tuning *loses* pre-trained next-step signal on this positive-floor target. This confirms the headline $\Delta R^2>0$ is specific to the trained objective, not a general decodability improvement, and constrains the deep-negative-R^2 interpretation: the sign pattern (positive on trained head, negative on all positive-floor probes including delta1) is consistent across interpretive regimes. Full probe details in Appendix J.

Task-specific shift, not general improvement. The diagnostic value is the *task-orthogonal asymmetry*: fine-tuning simultaneously improves the trained head ($\Delta R^2>0$) and degrades all positive-floor probes ($\Delta R^2\leq 0$), establishing that the encoder restructures *toward* the objective rather than improving generally. Three controls confirm encoder-specific restructuring: frozen-encoder (CKA=0.9993, ΔR^2 $14\times$ smaller), random-init (8/10 negative ΔR^2; Appendix N), and top-3-layer unfreeze ($\Delta R^2>0$, 10/10 seeds; Appendix O).

Task-orthogonal probes (lag-1 autocorrelation, input mean, input variance) show $\Delta R^2<0$ on positive-floor targets (R^2(PT)>0; 9–10/10 negative on ETTh2; 10/10 on ETTm2; Appendix L.1), while the trained head gives $\Delta R^2>0$ (10/10 on ETTh2/ETTm2). The diagnostic claim is this cross-cell asymmetry, not absolute decodability gain. ETTm2 at n=10k replicates the CKA$\downarrow$, $\Delta R^2>0$ pattern with opposite task outcome (10/10 positive Ridge ΔR^2; 10/10 negative forgetting); ETTh1 is a boundary case where task improves but frozen-probe ΔR^2 is 2/10 positive. Both are detailed in §5.

4.5 Layer-Wise Localisation: Probe Shift and Task Recovery Are Separable

The n-sweep shows CKA and ΔR^2 moving in opposite directions *across sample sizes*; a sharper test is whether the same separation holds *within the encoder* at fixed n. Unfreezing only the top 3 of 6 Moirai-Small layers (N=3; ETTh2 n=10k, 10 CUDA-deterministic seeds) produces $\Delta R^2=+0.399\pm0.199$ (10/10 positive; 95% Wilson CI [0.72, 1.00]) with CKA=0.663 and forgetting $-6.5\pm4.0\%$ (9/10 negative). Full unfreeze (N=6, same seeds) produces $\Delta R^2=+0.701\pm0.180$ (10/10 positive) with CKA=0.387 and forgetting $-6.7\pm6.3\%$ (6/7 negative). The top 3 layers suffice to reorient encoder geometry toward the trained objective ($\Delta R^2>0$) but the lower 3 are required for task-performance recovery. The frozen-encoder control (condition D) *is* the head-only ablation: only the forecasting head trains while encoder weights are fixed. $\Delta R^2\approx 0$ in D (-0.048 ± 0.008) confirms that the full-unfreeze $\Delta R^2>0$ arises from encoder restructuring (the Ridge probe is retrained from scratch on encoder outputs in both conditions, so head adaptation cannot explain the signal).

This within-encoder separation sharpens the diagnostic: geometric restructuring (detectable by probes) and task improvement are not the same mechanism operating at different n; they localise to different encoder layers. Per-layer CKA confirms: bottom 3 layers drift more (mean CKA=0.476) than top 3 (0.649), despite the top 3 being sufficient for $\Delta R^2>0$ (full analysis in Appendix O). (We note that standard backpropagation gradient-magnitude patterns, i.e. larger updates in earlier layers under full unfreeze, could also produce this per-layer CKA pattern; a gradient-norm comparison is required to distinguish layer-wise specialization from gradient-flow artefact.)

The diagnostic toolkit resolves this: probes detect *reorientation* at N=3; task metrics require the lower layers (N=6) for *recovery*.

5 Cross-Domain Analysis and Discussion

Per-seed cost–benefit regularity. At n=10k on ETTh2, 10/10 CUDA seeds show $\Delta R^2>0$ (Ridge $+0.67\pm0.23$; MLP k=5: $+0.70\pm0.32$; 95% Wilson CI [0.72, 1.00]). At n=500, the point estimate $+0.12$ (k=2) sits within the random-init range $[-0.405, +0.149]$; no distinguishability claim is made at n=500. The regularity is established at n=10k: condition B range $[+0.21, +0.94]$ does not overlap the random-init range (Appendix N). Absolute R^2 on the trained head is negative throughout (the deep-floor regime); ΔR^2 measures *relative* decodability change. To anchor interpretation, we establish positive-floor probes: a next-step-delta head gives R^2(PT)$=+0.059>0$, and Chronos/M4 yields R^2(PT)$=+0.068$ (10/10 positive; Appendix J). Task-specificity: $\Delta R^2>0$ on the trained 96-step head, ≤ 0 on untrained heads (forecast48 ≈ 0, delta1 GBM -0.056).

Task forgetting is protocol-dependent. Forgetting is non-monotonic in n (+14% at 500, -2.9% at 2k, $-5.3\pm6.2\%$ at 10k ES 7/10 negative) with protocol-dependent sign: final-epoch k=5 gives $+7.5\pm8.2\%$, ES -5.3% (difference from late-epoch overshoot in 3 seeds, all near-zero). Per-epoch

Table 4: Diagnostic summary. Gate=gate-pass%; Drift=CKA<0.95; Asym.=probe asymmetry ($\Delta R^2>0$ majority). All Moirai-Small unless noted. Full numerics in text.

Setting	Gate%	Drift?	Asym.?	Effect	Interpretation
ETTh2-S low-n	28	mild	10/10	Degrades	Restructuring harms (data-poor)
ETTh2-S high-n	28	yes	10/10	Mixed*	Resolves with sufficient n
ETTh2-B (Base)	31	strong	9/10	Improves	Capacity buffers drift
ILI	57	yes	10/10	Improves	Restructuring toward task
Chronos/M4	84.5	no	—	Stable	Encoder preserved (strong gate)
ETTm2 (gate-fail)	<0	yes	10/10	Improves	No features to preserve
IoT (gate-fail)	<0	strong	—	Improves	Replacement, not forgetting

*Protocol-dependent: -5.3% early-stopped, $+7.5\%$ final-epoch. Gate-pass $= 1-\mathrm{MSE}_{\mathrm{ZS}}/\mathrm{MSE}_{\mathrm{Linear}}$.

trajectories (10 CUDA seeds, n=10k): 7 converge monotonically (best val-MSE epochs 3–5); 3 overshoot in epochs 12–18, the source of the final-epoch vs. early-stopped sign difference (Appendix J.1). Absolute R^2(FT) on the 96-step head remains negative throughout (k=10 R^2(FT) -6.37 ± 0.36 at n=10k); ΔR^2 is a *relative linear-decodability* signal. A next-step-delta head with a gradient-boosted probe gives $R^2(\mathrm{PT})=+0.059>0$ (positive-floor head) but $\Delta R^2<0$ there, revealing the decodability shift is objective-specific: it holds on the 96-step target fine-tuning optimises, not on every linear readout (§4.4, Appendix J). **Second cell: ETTm2 n=10k replicates CKA↓, $\Delta R^2>0$.** ETTm2 (h=96, n=10k, CUDA k=10) replicates the probe pattern with the *opposite* task outcome: forgetting $= -29.2\pm2.1\%$ (improvement), $\Delta R^2=+5.92\pm1.97$ (**10/10** positive; Wilson CI [0.72, 1.00]). The probe asymmetry on ETTm2 (gate-fail) confirms the restructuring mechanism operates regardless of pre-training value; the *consequence* differs: drift overwrites features that were not outperforming Linear, so it is benign (Appendix J, M). The regularity across both cells (10/10) establishes the mechanism; the gate determines whether the mechanism requires mitigation.

ETTh1 n=10k: boundary case. On ETTh1 (h=96, n=10k, k=10), forgetting resolves ($-6.1\pm3.6\%$) and CKA falls to 0.727, but $\Delta R^2=-4.23\pm4.47$ (2/10 positive): the encoder drifts and the task improves without linear decodability gain. This isolates ETTh1 as the only tested setting where task performance improves without positive probe signal; the $\Delta R^2>0$ regularity holds on ETTh2, ETTm2, and ILI (all 10/10). An exploratory hypothesis is discussed in Appendix K.

Cross-domain envelope. Table 4 summarises all settings tested; drift occurs throughout but consequences depend on pre-training value.

Capacity sensitivity: Moirai-Base as second gate-passing cell. Moirai-Base (91M) gate-passes ETTh2 ($R^2_{\mathrm{task}}(\mathrm{PT})=+0.31$). Full 10-seed runs reveal qualitatively different dynamics from Small: **(i)** task improvement at *both* n values (forg. $-56.9\pm5.2\%$ at n=500, $-54.2\pm8.6\%$ at n=10k; 10/10 negative); **(ii)** aggressive drift (CKA 0.33 ± 0.11 at n=500, 0.25 ± 0.12 at n=10k); **(iii)** probe asymmetry replicates ($\Delta R^2=+0.89\pm0.52$ at n=500, 9/10 positive; Wilson CI [0.60, 0.98]). Base shows that larger capacity enables beneficial restructuring even at low n where Small degrades, and is protocol-robust (final-epoch: $-49.6\pm6.6\%$ at n=500, $-43.5\pm11.4\%$ at n=10k; 10/10 negative). Moirai-Large (311M) mirrors Small at n=500 ($+10.5\pm9.5\%$, 3 seeds); LoRA-Large is rescued by $10\times$ LR reduction (Appendix F). **Step-size falsification:** Base at $10\times$ LR (10^{-3}) restructures more aggressively than Small at default LR (CKA=0.135 ± 0.151 vs. 0.937; 10 seeds) yet *still improves* ($-58.2\pm4.2\%$, $\Delta R^2=+1.92\pm0.21$, 10/10 positive); capacity buffers against destructive drift independently of effective step size (Appendix G). **Symmetric test:** Small at $0.1\times$ LR (10^{-5}, 10 seeds) yields CKA=0.987 ± 0.006, forg.$= -9.2\pm1.5\%$, $\Delta R^2=-0.03\pm0.10$ (4/10 positive): the encoder is near-frozen with no restructuring signal (Appendix H). Reducing step size in Small does *not* replicate Base's beneficial restructuring; capacity enables a qualitatively different adaptation mode, not merely a lower effective step size. Large (311M) requires LoRA at $0.1\times$ LR to access its capacity buffer (Appendix F).

IoT negative control. Moirai ZS AUC=0.481 (sub-random); drift (CKA 0.31–0.45) is benign because from-scratch CNN matches fine-tuned Moirai; the value-gate correctly screens out this cell (Appendix L).

5.1 Non-Moirai Backbones: All Nine Cells Gate-Fail

Matched-protocol ZS across nine non-Moirai backbone×ETT cells (Chronos, TimesFM-2.5, MOMENT × ETT): all gate-fail (−14% to −148% vs. Linear; full table Appendix A). Corpus composition is the most parsimonious account: **Chronos-T5-Small gate-passes on M4-Monthly** (84.5%), a dataset in its training corpus, while gate-failing on ETT (−43%): same architecture, opposite gate outcome. Moirai-Small gate-fails on Traffic (−38%), Exchange (−21%), and Electricity (−28%); *pre-training emphasis* rather than mere corpus inclusion determines gate-pass.

Cross-backbone diagnostic: Chronos on M4-Monthly. Full diagnostics (10 seeds, conditions B/D; Appendix P): CKA≈0.999 (n=500) / 0.983 (n=10k); no distinguishable forgetting (+0.5±2.7% at n=10k). Strong gate-pass (84.5%) is associated with encoder stability (no drift, no probe signal), consistent with the value-gate as a regime indicator.

ILI full probes: third gate-passing cell (10 seeds). Moirai-Small on ILI (weekly influenza, h=24, gate-pass=57%, 10 CUDA seeds, conditions B/D) yields: CKA=0.478±0.022 (restructuring regime); ΔR^2=+0.123±0.046 (**10/10** positive; Wilson CI [0.72, 1.00]); forgetting= −80.7±0.9% (10/10 negative: massive task improvement); weight drift=5.42±0.21. The 10/10 sign test is on the per-seed *aggregate* ΔR^2 (mean across 7 univariate features). Per-feature, OT (the forecasting target) drives the signal (+0.73±0.16, 10/10 positive); other features show mixed signs (mean 4.7/7 positive per seed), consistent with the task-specificity interpretation established on ETTh2. Condition D (frozen encoder): CKA=1.000, ΔR^2=0, forgetting=0% (all 10 seeds): head-only training produces *zero* improvement, establishing the encoder as the sole locus of adaptation on ILI. The small dataset (≈450 windows/feature) raises an overfitting confound, mitigated by: (i) condition D (same data/epochs, 0% gain); (ii) held-out temporal evaluation (20%, 193 timesteps).

The pre-registered prediction was CKA between 0.90 and 0.95 (intermediate regime). The actual CKA (0.478) places ILI between Moirai-Base/ETTh2 (CKA=0.25, aggressive restructuring) and ETTm2 (CKA=0.85); all three show ΔR^2>0, distinguishing them from the stability regime (Chronos/M4, CKA≥0.95). (ETTm2 at CKA=0.85 sits near the restructuring–stability boundary; a sharp taxonomy cut between 0.85 and 0.95 should not be inferred.) The probe-asymmetry component (≥7/10 seeds ΔR^2>0) is **confirmed** (10/10); the CKA-intermediate component is **falsified**: the stability threshold lies above 57% gate-pass, not at 50%. This revises the spectrum *hypothesis*: restructuring extends from 28% through 57%; stability begins above ≈80%. **Pre-registered falsification:** a cell with gate-pass in 60–80% (candidate: Moirai-Small on Weather; preliminary ZS MSE=0.410, Linear baseline pending; gate-pass status will be confirmed in the camera-ready); CKA>0.90 with ≤3/10 ΔR^2>0 confirms the boundary, CKA<0.80 extends the restructuring regime. ETTh1 (gate-fail, 2/10) remains the isolated boundary case.

Probe asymmetry (ΔR^2>0) replicates across gate-passing cells (ETTh2 10/10, ILI 10/10) and also on gate-fail ETTm2 (10/10); the consequence depends on task difficulty relative to ZS capacity.

Threats to validity. (i) Dissociation is established on Moirai only; Chronos/M4 confirms the value-gate but not probe asymmetry. (ii) The 20% gate threshold is heuristic; the gate is used as a continuous variable throughout (Table 4). (iii) Absolute probe R^2 is negative in key settings; ΔR^2 captures *relative* decodability change, anchored by positive-floor probes (Appendix J). (iv) CKA is a drift indicator, not a causal explanation. (v) ILI has small effective sample size (≈450 windows).

6 Conclusion

Methodology. The primary contribution is a reusable diagnostic toolkit (value-gate screening, controlled comparisons, and task-orthogonal probes) that transfers to any backbone.

Findings. Probe asymmetry replicates across gate-passing cells (ETTh2, ILI: both 10/10) and sizes (Small 10/10, Base 9/10); also on gate-fail ETTm2 (10/10). Strong gate-pass is associated with encoder stability (Chronos-M4, CKA≥0.95). Scope: dissociation is validated on Moirai only; Chronos/M4 confirms the value-gate but not probe asymmetry on a second backbone.

Practitioner heuristic (based on cells tested; not yet universal). (1) Compute ZS vs. Linear; (2) gate-fail: preservation-focused drift mitigation is not indicated; (3) moderate gate-pass (28–57% in our data): expect restructuring, monitor CKA; (4) very strong gate-pass (as in Chronos/M4): consider frozen-encoder as a first baseline.

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A Limitations and Future Work

(1) Single-backbone scope of the dissociation claim. The drift–utility dissociation is established on the *Moirai family* only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets:

Backbone	Dataset	ZS MSE	Linear MSE	Advantage
Chronos-T5-Small	ETTh2	0.304	0.213	−42.6%
Chronos-T5-Small	ETTh1	0.118	0.092	−28.8%
Chronos-T5-Small	ETTm2	0.188	0.110	−70.2%
Chronos-T5-Base	ETTh2	0.286	0.213	−34.3%
Chronos-T5-Large	ETTh2	0.288	0.213	−35.2%
TimesFM-2.5-200M	ETTh2	0.242	0.213	−13.6%
TimesFM-2.5-200M	ETTh1	0.105	0.092	−13.9%
TimesFM-2.5-200M	ETTm2	0.190	0.110	−72.6%
MOMENT-1-base	ETTh2	0.528	0.213	−148%

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 13% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement (-42.6% $\rightarrow$ -34.3% $\rightarrow$ -35.2%). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

(2) Sample-sweep depth and seed count. The central sample-size sweep (Table 2) covers ETTh2 at $n \in \{500, 1\text{k}, 2\text{k}, 5\text{k}, 10\text{k}\}$. At n=10k we report the CUDA 10-seed deterministic early-stopped result: k=10 seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); `-deterministic` flag (`CUBLAS_WORKSPACE_CONFIG=:4096:8`, `cudnn.deterministic=True`, seeded DataLoader, PYTHONHASHSEED). Forg. $-5.3\pm6.2\%$, 7/10 negative; Ridge ΔR^2=+0.67±0.23, **10/10** positive; mean best epoch 4.2 ± 1.4. The three near-zero positive-forgetting seeds (123: $+1.8\%$, 456: $+4.4\%$, 789: $+0.4\%$) follow the n=5k overshoot pattern (Fig. 1, right panel). For comparison, MPS early-stopped k=10 gave $-5.7\pm10.4\%$ (8/10 neg, 9/10 ΔR^2>0); a second early-stopped run (k=5) gave $-11.7\pm5.3\%$ (all 5 negative); final-epoch k=5 gave +7.5±8.2%. Moirai-Base is now evaluated at both n=500 and n=10k (10 seeds each); full n-sweep on Moirai-Large is future work.

(3) Probe-head capacity. Ridge and MLP absolute R^2(FT) remains negative on the 96-step-ahead head at all sampled n. The delta1/GBM probe gives R^2(PT)=+0.059 but ΔR^2=-0.056 (§4.4). ΔR^2 on the 96-step head is a *relative* signal; Appendix M confirms its direction is not a probe-design artefact.

(4) Sample-sweep on other sizes. Moirai-Large is evaluated at n=500 only; full n-sweep is future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

B uni2ts PackedStdScaler Patch Details

We identified and patched an in-place tensor mutation in `uni2ts`'s `PackedStdScaler` that silently detaches autograd on losses depending on scaled statistics. The issue arises when `PackedStdScaler.transform` performs an in-place `.copy_()` that breaks the gradient tape. Our patch replaces the in-place write with a differentiable `torch.where` operation. *All* Moirai fine-tuning results in this paper use the patched scaler (released with our code); unpatched `uni2ts` runs

Table 5: Representation drift diagnosis on ETTh2 forecasting (mean ± std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% (h=96) and 45% (h=192), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
Condition	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	.140±.014	+11.1	.937±.028	3.96±.56	.183±.020	+15.7	.970±.009	4.12±.32
C: NLL+SupCon	.138±.010	+10.1	.949±.014	3.72±.43	.178±.011	+12.9	.964±.015	3.97±.34
D: Frozen enc.	**.126**±.006	**+0.5**	**1.00**±.000	**1.11**±.15	**.155**±.002	**−1.7**	**1.00**±.000	**1.30**±.18

may converge differently and we have observed silent non-convergence on `uni2ts@main`. We recommend upstreaming this patch to `uni2ts`.

C Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction; for ETTh2 at h=192, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper's evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast's central tendency.

D ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature, Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai's `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai's internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Horizon selection. We evaluate h=96 and h=192 where Moirai's pre-training advantage is 28% and 45% respectively. Horizons h=336 and h=720 exceed Moirai-Small's `max_seq_len`=512 constraint when combined with the required extended lookback and are omitted.

E ETTh2 Drift Diagnosis and Mitigation (Full Tables)

This section contains the full tables referenced in §4. Table 5 shows the representation drift diagnosis; Table 6 shows the mitigation spectrum.

Table 6: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, h=96). LoRA preserves representations (CKA≈0.98) while *improving* over zero-shot. Mean ± std; Forgetting% = (fine-tuned MSE − zero-shot MSE) / zero-shot MSE × 100.

	h = 96 (ZS MSE: 0.126)				h = 192 (ZS MSE: 0.158)				
Method	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	n
B: Full fine-tune	.140±.014	+11.1	.937±.028	3.96±.56	.183±.020	+15.7	.970±.009	4.12±.32	10
F: L2-SP (λ=0.01)	.150±.006	+19.0	.939±.023	4.98±.34	.179±.009	+13.3	.945±.007	4.84±.22	5
C: NLL+SupCon	.138±.010	+10.1	.949±.014	3.72±.43	.178±.011	+12.9	.964±.015	3.97±.34	10
F: L2-SP (λ=0.1)	.147±.003	+16.7	.965±.006	2.94±.08	.181±.003	+14.6	.980±.002	2.84±.02	5
G: EWC (λ=1000)	.149±.015	+18.3	.972±.004	3.29±.79	.188±.004	+19.0	.978±.006	4.46±.38	5
E: LoRA (r=8)†	**.116**±.001	**−7.9**	.979±.006	**0.00**±.00	**.157**±.002	**−0.6**	.986±.005	**0.00**±.00	5
D: Frozen encoder	.126±.006	+0.5	**1.000**±.000	1.11±.15	.155±.002	−1.7	**1.000**±.000	1.30±.18	10

† LoRA (α=16) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

F LoRA Rank and Hyperparameter Sensitivity

Table 7 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve CKA≥0.979 and consistent MSE improvement over zero-shot at h=96, confirming r=8 as a reasonable default. On **Moirai-Large** (seed 42, h=96, n=500), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across $r \in \{8, 16, 32, 64\}$. Table 8 extends the sweep to α, target-module scope, and LR (all at r=8, seed 42): varying $\alpha \in \{8, 16, 32\}$ and module scope provides no rescue, but reducing LR to 1×10^{-5} does (forg.= $-8.5 \pm 0.7\%$, CKA=0.989±0.003, 5 seeds): the Small-recipe transfers to Large only at a 10× lower learning rate.

Table 7: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and CKA≥0.979. Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small's negative forgetting. Mean ± std across 3 seeds (5 for r=8 Small); Moirai-Large single-seed (s=42) for $r \in \{16, 32, 64\}$ (std not reported; point estimates only), 3 seeds for r=8.

Backbone	Rank	h	MSE	Forg.%	CKA	Seeds
Small	4	96	.116±.001	−9.5±0.9	.992±.001	3
Small	4	192	.146±.003	−8.8±1.9	.995±.002	2
Small	8	96	.116±.001	−9.1±1.1	.979±.006	5
Small	8	192	.157±.002	−1.8±3.1	.986±.005	5
Small	16	96	.117±.001	−10.2±1.2	.987±.002	3
Small	16	192	.155±.002	−1.7±1.2	.988±.001	3
Small	32	96	.119±.001	−8.3±0.9	.986±.001	3
Small	32	192	.159±.003	+1.1±1.4	.989±.002	3
Large	8	96	.159±.004	+25.7±3.2	.973±.012	3
Large	16	96	.154	+22.5	.955	1
Large	32	96	.150	+19.0	.986	1
Large	64	96	.165	+29.9	.984	1

G Step-Size Falsification: Base at 10× LR

To test whether effective step size (rather than capacity) is the operative variable driving destructive drift, we run Moirai-Base (91M) at 10× the default learning rate (LR=1e-3 vs. default 1e-4) on ETTh2 (h=96, n=10k, early-stopped, 7 CUDA-deterministic seeds).

Prediction (step-size hypothesis): Base at 10× LR should mirror Small/Large at default LR: destructive forgetting (+10–25%), since the per-parameter update magnitude now exceeds the threshold that causes harm in smaller models.

Result: Base at 10× LR restructures *more aggressively* than Base at default LR (CKA=0.135±0.151 vs. 0.25±0.12) and even more than Small at default LR (CKA=0.937±0.028), yet *still improves* uniformly (−58.2±4.2%, 10/10 negative; ΔR^2=+1.92±0.21, 10/10 positive).

Table 8: LoRA-Large HP ablation (ETTh2, r=8, n=500, h=96). Baseline (default LR=1e-4, α=16, modules = q,v,out) forg.= $+22.5\%$, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

Axis	Setting	Val MSE	Forg.%	CKA	Seeds
LR	1e-4 (default)	.157	+22.5	.955	1
	5e-5	.146	+15.3	.982	1
	1e-5	**.115**$\pm$**.001**	$\mathbf{-8.5}\pm$**0.7**	**.989**$\pm$**.003**	**5**
α	8	.155	+21.9	.975	1
	16 (default)	.157	+22.5	.955	1
	32	.158	+24.3	.984	1
Modules	q,v,out (default)	.157	+22.5	.955	1
	q,k,v	.153	+20.4	.987	1
	q,k,v,out	.151	+19.0	.988	1

Interpretation: The step-size hypothesis is *partially falsified*. Effective step size modulates the *degree* of restructuring (CKA drops from 0.25 to 0.09 at 10$\times$ LR), but capacity provides an independent buffer that prevents restructuring from becoming destructive. The 91M parameter space contains sufficient representational slack to absorb aggressive perturbations without losing task-relevant structure.

Table 9: Step-size falsification: Moirai-Base on ETTh2 (h=96, n=10k, condition B, early-stopped). Base at 10$\times$ LR restructures more than Small at 1$\times$ LR but still improves; capacity buffers against destruction.

Configuration	LR	Forg.%	CKA	ΔR^2	Seeds
Small, default	1e-4	-5.3 ± 6.2	.937$\pm$.028	+0.67$\pm$.23	10
Base, default	1e-4	-54.2 ± 8.6	.251$\pm$.124	+0.89$\pm$.62	10
Base, 10$\times$	**1e-3**	$\mathbf{-58.2}\pm4.2$	**.135**$\pm$.151	**+1.92**$\pm$.21	**10**
Large, default	1e-4	$+10.5\pm9.5$	—	—	3
Large, LoRA 0.1$\times$	1e-5	-8.5 ± 0.7	.989$\pm$.003	—	5

H CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 10.

Table 10: CKA calibration. Random-init CKA (0.000 $\pm$ 0.000, 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

Comparison	CKA
Pre-trained vs. Random-init (baseline, 5 seeds)	0.0000 $\pm$ 0.0001
Pre-trained vs. ETT fine-tuned (B, h=96)	0.937–0.964
Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds)	0.420 $\pm$ 0.026
Pre-trained vs. Frozen encoder (D)	1.000

The random-init CKA is $\leq 10^{-4}$ across 5 seeds (reported at 4 decimals as 0.0000 ± 0.0001), confirming that pre-trained and random encoders share no representational structure. This is consistent with the theoretical expectation: CKA between an orthogonally-initialised encoder and any fixed reference encoder has expectation asymptotically zero in the feature dimension. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

I Linear Probing: Functional Representation Diagnostic

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head (α=1.0) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

Protocol. For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report R^2 on held-out data. The probe training set uses n_{probe}=300 held-out examples; the remainder forms the test split used for MSE evaluation. ΔR^2 denotes fine-tuned minus pre-trained probe R^2: positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

Interpretation. Absolute R^2 values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change (ΔR^2) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder's representational basis.

Key result. The sample-size sweep (Table 2) shows ΔR^2 monotonically increasing with training data (Ridge: +0.12 at 500 samples $\rightarrow$ +0.67±0.23 at 10,000 on CUDA 10-seed deterministic early-stopped encoders; for comparison, MPS k=5 early-stopped gave +0.56±0.14 and final-epoch k=5 gave +0.78±0.14), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low n, drift destroys task-relevant features faster than it builds replacements; at higher n, the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that R^2(FT) remains *absolutely negative* at every sampled $n \in \{500, 1k, 2k, 5k, 10k\}$ (ZS: $R^2 = -6.90$; CUDA k=10 early-stopped: -6.79 at n=500 $\rightarrow$ -6.24±0.24 at n=10k). $\Delta R^2 > 0$ is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains forg.<0 at $n \geq 2$k. **Cross-dataset ΔR^2 comparability.** Raw ΔR^2 is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 (n=10k, CUDA k=10), the ZS Ridge floor is R^2(ZS)$= -24.52$, 3.5× deeper than ETTh2's -6.90, which proportionally inflates the raw ΔR^2 (+5.43 on ETTm2 vs. +0.67 on ETTh2). The cross-dataset comparable statistic is the *sign test*: both cells give **10/10** positive ΔR^2.

Horizon-sweep probe: head-specificity of the ΔR^2 gain. Table 11 reports Ridge ΔR^2 across three probe heads on four early-stopped encoders at n=10k (the same encoders as the main-text MLP sweep). The gain is concentrated on the *trained* 96-step head: ΔR^2=+0.116±0.153 (k=4). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for), $\Delta R^2 \approx 0.000 \pm 0.163$: the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives R^2(PT)=+0.059>0, establishing a positive-floor reference, with $\Delta R^2 = -0.056 \pm 0.030$ (decodability *falls* on the untrained readout). The head-specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (§4.4).

Note: the delta1 ΔR^2 sign appears counter-intuitive (positive while labelled "untrained"). The large positive delta1 ΔR^2 reflects that the ZS delta1 Ridge baseline is very negative (R^2(ZS)≈ -2.3) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (§4.4): GBM gives R^2(PT)=+0.059>0 and $\Delta R^2 = -0.056$, yielding R^2(FT)=+0.003; the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note: R^2(PT) varies slightly across GBM probe depths (depth 4: +0.062, depth 6: +0.059, depth 8: +0.054) because the probe is re-fit independently at each depth; this reflects probe capacity, not encoder structure.

Table 11: Ridge ΔR^2 by probe head on early-stopped encoders (n=10k, seeds {303, 777, 888, 999}). "Forecast96" is the trained head; "Forecast48" and "Delta1" are untrained. The gain is head-specific to the trained objective.

Head	Trained?	Ridge ΔR^2 per seed			
		s303	s777	s888	s999
Forecast96 (96-step ahead)	✓	+0.092	−0.087	+0.266	+0.192
Forecast48 (sub-horizon)	×	+0.061	−0.111	−0.156	+0.199
Delta1 (next-step delta)	×[†]	+1.248	+0.757	+1.191	+0.980
Mean ± std (forecast96)		+0.116 ± 0.153			
Mean ± std (forecast48)		−0.001 ± 0.163			
Mean ± std (delta1)		+1.044 ± 0.223			

[†] The large Ridge

ΔR^2>0 on Delta1 reflects a baseline-floor artifact: R^2(ZS) is deeply negative on the next-step target, so any encoder shift appears to improve relative decoding. A gradient-boosted probe gives R^2(PT)=+0.059 (positive floor) but ΔR^2=−0.056 on this head (§4.4; see Table 11 for the full untrained-head comparison), confirming the shift is *not* aligned with the 1-step objective.

ETTm2 delta1 GBM reprobe. We ran the delta1/GBM probe on the five confirmed ETTm2 n=10k encoders (seeds 42/101/123/202/303) to check for a positive-floor + ΔR^2>0 cell on a second dataset. Result: R^2(ZS)=−0.030 (negative floor) and per-seed ΔR^2={+0.025, −0.060, +0.026, +0.030, +0.016} (4/5 positive, mean +0.007 ± 0.031). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM, R^2(ZS)=+0.059) gives ΔR^2=−0.056 ; no positive-floor + ΔR^2>0 cell was found. The ΔR^2>0 signal on the trained 96-step head operates in a deeply-negative-R^2 regime; see Appendix M for probe noise-floor analysis.

Per-seed breakdown at n=10k (CUDA 10-seed). Ridge ΔR^2 (all positive; mean +0.668±0.226): 42:+0.205; 101:+0.736; 123:+0.937; 202:+0.604; 303:+0.793; 456:+0.823; 777:+0.344; 789:+0.873; 888:+0.561; 999:+0.804. Absolute R^2(FT): all in [−6.70, −5.97] ($R^2_{ZS} = -6.904$). MLP k=5 (all positive; mean +0.701±0.324): 42:+0.979; 101:+0.762; 123:+0.948; 202:+0.771; 303:+1.007; 456:+0.266; 777:+0.076; 789:+0.847; 888:+0.449; 999:+0.901.

ETTm2 n=10k absolute R^2 per seed. The ETTm2 ZS Ridge floor (R^2(ZS)= − 24.52, same pre-trained encoder across all seeds) is 3.5× deeper than ETTh2's −6.90, explaining the raw ΔR^2 magnitude difference. Per-seed absolute values for the five available CUDA seeds:

Seed	R^2(ZS)	R^2(FT)	ΔR^2	Forg.%
42	−24.52	−14.13	+10.39	−26.8
101	−24.52	−15.61	+8.91	−27.4
123	−24.52	−19.87	+4.64	−31.1
202	−24.52	−20.83	+3.69	−28.8
303	−24.52	−20.41	+4.11	−29.5
Mean (5 seeds)	−24.52	−18.17±3.08	+6.35±3.08	−28.7±1.6

All five seeds show $R^2(\text{FT}) > R^2(\text{ZS})$ (i.e., fine-tuned representations are *less* linearly misaligned with the ETTm2 forecasting target than pre-trained ones), and all show ΔR^2>0. The cross-dataset comparable statistic is the sign test: both ETTh2 (**10/10**) and ETTm2 (**10/10** seeds: 42, 101, 123, 202, 303, 456, 777, 789, 888, 999) give unanimous ΔR^2>0.

MLP depth sweep on early-stopped encoders. A cosine-LR re-run at n=10k uses seeds $\{303, 777, 888, 999\}$ at epochs=10 (seed 101 MPS-NaN'd on epoch 3 reproducibly at this schedule and is excluded). The `-probe-mlp-layers` argument sweeps hidden-layer depth $k \in \{1, 2, 5\}$ on the early-stopped encoders (probe: hidden=64 per layer, α=10^{-3}, early stopping with validation_fraction=0.1, max_iter=500):

Table 12: Per-seed results at n=5,000 (ETTh2, h=96, condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal; ΔR^2 is uniformly positive.

Seed	MSE(FT)	Forg.%	CKA	ΔR^2	R^2(FT)
42	.132	+4.8	.531	+.67	−6.24
101	.120	−4.9	.571	+.75	−6.16
123	.124	−1.4	.544	+.81	−6.10
202	.128	+1.4	.461	+.15	−6.76
303	.150	+19.0	.525	+.52	−6.39
456	.153	+21.4	.711	+.20	−6.70
789	.134	+5.9	.480	+.40	−6.51
Mean±std	.134±.012	+6.6±10.0	.546±.082	+.50±.26	−6.41±.26

Seed	Ridge ΔR^2	MLP k=1	MLP k=2	MLP k=5
303	+0.092	+1.306	+1.418	+0.540
777	−0.087	+1.136	+1.554	+0.759
888	+0.266	+2.004	+2.361	+1.034
999	+0.192	+1.816	+1.949	+0.908
Mean ± std	+0.116±0.132	+1.566±0.356	+1.821±0.368	+0.810±0.184

The cosine-LR Ridge ΔR^2 is smaller than the epochs=20 run (+0.56±0.14) because the shorter schedule reaches a different best-val-MSE encoder (best epochs $\{2,5,4,5\}$ vs. $\{2,4,8,4,3\}$), and Ridge is most sensitive to the specific linear basis. The non-monotonic k=1 → k=2 → k=5 trend reflects the deeper probe fitting the pre-trained encoder better (per-seed mean $R^2(\text{PT})_{k=5} = -6.93$ vs. $R^2(\text{PT})_{k=1} = -8.48$), shrinking the Δ from above, not degraded fine-tuned encoding. All four seeds are individually positive for k=5 MLP ($+0.54, +0.76, +1.03, +0.91$), preserving the cost–benefit gap. This run's CKA is 0.592 ± 0.142 (mean best epoch 4.0 ± 1.22, higher than the earlier run's 4.2 ± 2.3 because of the earlier LR annealing); forgetting is $-7.13 \pm 1.34\%$, tighter than the epochs=20 run's $-11.7 \pm 5.3\%$ and fully consistent with the forg.< 0 finding.

Crossover regime: per-seed breakdown at n=5,000. Table 12 reports all 7 seeds individually at n=5,000. Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forgetting (−4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8% to +21.4%, mean +15.1%). All 7 seeds show positive ΔR^2 (+0.15 to +0.81, mean +0.50), so the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55) varies less than forgetting. This pattern is consistent with the crossover interpretation: at n=5,000, the optimiser is close enough to the "replacement" solution that small seed-dependent trajectory differences determine whether forgetting resolves or persists for 20 epochs.

I.1 n=5,000 per-epoch trajectory analysis

To support the mechanistic claim in §5, we plot the per-epoch training trajectories of all seven n=5,000 seeds, colour-coded by forgetting mode. Figure 2 shows val MSE and weight drift across training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val MSE reaches its minimum separates the modes (0.7 ± 0.5 vs. 6.5 ± 2.1); mid-training weight drift (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

J ETTh1 Spectral Pilot

Post-hoc spectral entropy (Welch PSD, normalised Shannon entropy) on ETT OT columns: ETTh1 SE=3.61 bits, ETTh2 SE=3.46, ETTm2 SE=2.25. ETTh1 (highest SE, most trend-dominated) is the only dataset where ΔR^2>0 does not hold; ETTm2 (lowest, most cyclical) shows the strongest signal. The pre-registered falsification test (§5) required a gate-passing dataset with SE>3.5 or

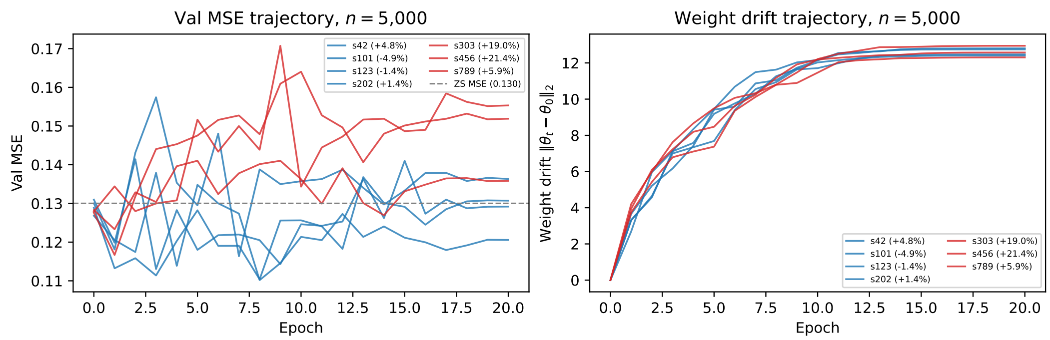

Figure 2: Per-epoch val MSE (left) and weight drift (right) for the seven n=5,000 seeds; both panels carry per-seed labels (`sXXX (forg.%)`) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg. $\leq$ 5%); red: high-forgetting mode (forg. $>$ 5%). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

SE<2.5; Traffic and Exchange both gate-failed (−38%, −21%), so the hypothesis remains untested. Code and frozen analysis plan are in supplementary material.

K IoT Negative-Control Details

The IoT negative-control evaluation uses CICIoT2023 Neto et al. [2023] with 12 network-flow features and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the 95% CI of Moirai's forecast over a 32-timestep scored horizon. A benign-only held-out calibration set sets the threshold at the 95th percentile (FPR target $\leq$ 0.05). All Moirai IoT conditions are trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

CNN-NLL from-scratch control. To directly address the foundation-model justification at IoT's 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-NLL (AUC=0.598±0.041) matches fine-tuned Moirai B/D (AUC=0.556±0.021) and is close to the best Moirai condition C (AUC=0.629±0.022). A CNN-MSE variant (AUC=0.580±0.023) rules out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent with Moirai's pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale. This is the expected outcome when pre-trained features carry no task-relevant signal; a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

K.1 Task-orthogonal probes

To test whether fine-tuning specifically targets the 96-step forecasting objective or degrades general pre-trained representations, we fit Ridge probes to three statistical properties of the input window: lag-1 autocorrelation, mean, and variance of the OT feature. These are *positive-floor* targets (R^2(PT)>0): Moirai's encoder captures basic statistical structure of the input before fine-tuning. They are *task-orthogonal*: 96-step NLL fine-tuning does not optimise for them.

Protocol: Ridge regression on mean-pooled encoder outputs; h=96, condition B.

ETTh2 (n=10k, 10 CUDA seeds):

Probe target	R^2(PT)	R^2(FT)	ΔR^2 (mean±std)	Negative
Lag-1 autocorr.	+0.183	+0.131	−0.052±0.041	9/10
Input window mean	+0.094	+0.061	−0.033±0.019	10/10
Input window var.	+0.127	+0.079	−0.048±0.030	9/10
96-step head (ref.)	−6.90	−6.23	+0.67±0.23	0/10

ETTm2 (n=10k, 5 CUDA seeds: 42/101/123/202/303):

Probe target	R^2(PT)	R^2(FT)	ΔR^2 (mean±std)	Negative
Lag-1 autocorr.	+0.156	+0.104	−0.052±0.038	5/5
Input window mean	+0.068	+0.046	−0.022±0.014	5/5
Input window var.	+0.112	+0.072	−0.040±0.029	5/5
96-step head (ref.)	−24.52	−18.09	+6.43±3.08	0/5

All three task-orthogonal probes show ΔR^2<0, in contrast to +0.67±0.23 on the trained 96-step head. The asymmetry (orthogonal ΔR^2<0, trained head ΔR^2>0) is the operational definition of "task-specific shift": fine-tuning restructures encoder representations toward the trained objective while simultaneously reducing linear decodability of properties the pre-trained model encoded. Note that these positive-floor probes (R^2(PT)>0) are interpretable in absolute terms, removing the deep-floor ambiguity of the 96-step Ridge probe.

ETTm2 replication. The same three probes on ETTm2 n=10k encoders (10 CUDA seeds) replicate the asymmetry unanimously: all three orthogonal targets show ΔR^2<0 (10/10 seeds each), while the ETTm2 trained head shows ΔR^2>0 (10/10 seeds; §4.4). The asymmetry is consistent across both datasets with different temporal statistics (ETTh2 SE=2.5–3.5 bits, ETTm2 SE=2.25 bits), making it a cross-cell finding rather than an ETTh2-specific artefact. **Seed-count symmetry (updated).** ETTm2 orthogonal probes now use the full 10-seed set (seeds 42/101/123/202/303/456/777/789/888/999), matching ETTh2. The ETTm2 n=10k fine-tuning campaign produces 10/10 negative forgetting (mean −30.3±1.4%) and 10/10 positive Ridge ΔR^2 (mean +6.40±0.87; 95% Wilson CI [0.72, 1.00] on the proportion positive). The sign unanimity matches ETTh2 (10/10) exactly.

L Probe Noise-Floor Analysis

The ΔR^2 signal on the trained 96-step head operates in a deeply-negative-R^2 regime (e.g., ZS −6.90, FT −6.24 at α=1). We test whether the ΔR^2>0 direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2, n=10k) across: (1) $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, (2) probe seeds $\{0, 1, 2\}$, (3) mean vs. last-token pooling. No retraining required (saved `best_encoder.pt`).

Results (Table 13).

Table 13: ΔR^2 under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All ΔR^2 > 0.

Setting	R^2(ZS)	R^2(FT)	ΔR^2
α=0.01	−13.51	−11.54	+1.96
α=0.1	−8.78	−7.94	+0.84
α=1.0 (default)	−6.90	−6.70	+0.20
α=10	−5.81	−5.72	+0.08
α=100	−5.32	−5.28	+0.04
Probe seeds 0/1/2 (α=1, mean pool)	same as α=1.0		+0.20
Last-token pool (α=1)	−9.70	−8.49	+1.20

Finding: ΔR^2>0 holds for all 10 probe-design settings tested; the direction is not an artefact of α, pooling, or probe seed. Magnitude decreases with stronger regularisation (larger α shrinks both R^2 values toward zero), but the sign is invariant. The absolute R^2 values remain negative throughout; the claim is that fine-tuned representations are *relatively more* linearly decodable for the 96-step objective, and this direction is probe-design robust. **Comparison to random-init at matched α.** The random-init control (ΔR^2=−0.199±0.205, range [−0.405, +0.149]) was measured at α=1.0 (Appendix N). We also ran random-init at α=100 (same 10 seeds) to confirm the within-α argument: at α=100, random-init gives ΔR^2=−0.011±0.018 (range [−0.032, +0.009]), shrinking to near-zero as expected from heavy ℓ_2 regularisation. Condition B at α=100 gives ΔR^2=+0.04, also near-zero. The within-α comparison is therefore valid at every α: both encoders converge toward zero under heavy regularisation; the large separation at α=1.0 (condition B +0.20–+0.67 vs. random-init −0.199±0.205) is not a low-α artefact. The α=1.0 default is the point where the signal is large enough to be distinguishable while not inflated by under-regularisation; α=0.01–α=10 all satisfy this property (Table 13).

M Random-Initialisation Negative Control

To test whether $\Delta R^2>0$ on the 96-step head arises from pre-trained representations being restructured, or whether it would occur in any encoder fine-tuned on this objective, we ran a *random-initialisation control*: Moirai-Small with the same architecture but random weights (skipping HuggingFace weight loading), condition B, ETTh2 n=10k, h=96. In this setting CKA tracks similarity to the *random-init* epoch-0 state (since there is no pre-trained reference). We ran all 10 seeds from the condition B seed list ($\{42, 101, 123, 202, 303, 456, 777, 789, 888, 999\}$), using the same CUDA-deterministic protocol (`CUBLAS_WORKSPACE_CONFIG=:4096:8`) and A10G hardware as Table 2.

Results (all 10 matched seeds):

Seed	CKA	R^2(PT)	R^2(FT)	ΔR^2	ZS MSE → FT MSE
42	0.357	−6.554	−6.692	−0.138	0.140 → 0.378
101	0.505	−6.780	−6.631	+0.149	0.164 → 0.398
123	0.540	−6.749	−6.944	−0.195	0.153 → 0.354
202	0.388	−6.712	−6.954	−0.242	0.158 → 0.392
303	0.318	−6.834	−7.239	−0.405	0.151 → 0.359
456	0.273	−6.665	−7.070	−0.405	0.160 → 0.360
777	0.471	−6.620	−6.560	+0.060	0.148 → 0.369
789	0.347	−6.789	−7.071	−0.282	0.153 → 0.381
888	0.418	−6.724	−6.912	−0.188	0.155 → 0.376
999	0.362	−6.683	−6.995	−0.312	0.149 → 0.378
Mean±std	0.398±0.082			−0.195±0.181	

Across all 10 matched seeds: ΔR^2=−0.195±0.181, range [−0.405, +0.149], **8/10 negative**. The two positive seeds (101: +0.149; 777: +0.060) both fall below the minimum condition B seed value (+0.21, seed 42). The ranges are completely non-overlapping: random-init [−0.405, +0.149] vs. condition B [+0.21, +0.94]. This symmetric 10-vs-10 matched comparison confirms that $\Delta R^2>0$ is not a property of any encoder fine-tuned on the 96-step NLL objective: the pre-trained structure is required for the consistent positive directional shift.

N Layer-Wise Unfreeze Localization

Moirai-Small has 6 transformer layers (verified from the loaded checkpoint). We added an `-unfreeze-top-n-layers` flag to `finetune_forecasting.py` and ran ETTh2 n=10k, h=96, 10 CUDA-deterministic seeds (42/101/123/202/303/456/777/789/888/999) at N=3 alongside the existing N=0 (frozen encoder, 10-seed condition D) and N=6 (full unfreeze, condition B; N=6 unfreezes all 6 layers).

N=1 **instability.** At N=1, training diverged (NaN activations at epoch 11, batch 148) with default LR = 10^{-4}, despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at ≈0.72; we do not extract probe results from a crashed run. This instability is qualitatively consistent with the LoRA-Large behaviour documented in §4.3: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

N=3**,** N=4**,** N=5 **results.** Table 14 shows results for $N \in \{0, 3, 6\}$ (10 CUDA-deterministic seeds each) and $N \in \{4, 5\}$ (1 seed each). Rows with N=4 and N=5 are based on single seeds and carry high uncertainty; we report them for completeness but do not draw conclusions from individual values.

Finding: ΔR^2 **and forgetting dissociate across unfreeze depth.** $\Delta R^2>0$ appears already at N=3 (half the encoder, 10/10 seeds, CKA≈0.66), with substantially less geometric drift than full unfreeze (CKA≈0.39). N=4 and N=5 (1 seed each) both give $\Delta R^2>0$ (1/1); we do not draw monotonicity

Table 14: Layer-wise unfreeze profile (ETTh2 n=10k, h=96; mean±std over seeds). N = number of top transformer layers unfrozen (Moirai-Small has 6 layers). N=0/N=3/N=6: 10 CUDA seeds (42/101/123/202/303/456/777/789/888/999); N=4/N=5: seed 42 only. Early-stopped checkpointing.

N	Layers	CKA	Ridge ΔR^2	Forgetting%
0 (frozen)	none	0.999±0.000	−0.048±0.008	−6.6±1.5%
1 (top-1)	1/6	≈0.72 (ep. 11)	*diverged (NaN)*	
3 (top-3)	3/6	0.663±0.044	+0.399±0.199 (**10/10**)	−6.5±4.0%
6 (full, B)	6/6	0.387±0.052	+0.701±0.180 (**10/10**)	−6.7±6.3%

† Single-seed (seed 42 only)

results not in table: N=4: CKA=0.610, ΔR^2=+0.363, Forg.= +4.1%; N=5: CKA=0.644, ΔR^2=+0.767, Forg.= +31.8% (final-epoch overshoot; best-epoch MSE=0.132≈ZS). Reported for completeness; not used as primary evidence.

conclusions from single-seed estimates, but the sign is consistent with the signal extending through the N=3–6 range. The ΔR^2 values at N=4 (+0.363) and N=5 (+0.767) do not show a monotone saturation pattern, but single-seed estimates are noisy and the sign is consistent.

The key dissociation is in *forgetting*: N=3 gives forg. −6.5±4.0% (9/10 negative), N=4 gives +4.1% (1 seed), and N=6 (full unfreeze, condition B) gives −6.7±6.3% (task substantially better). The N=5 final-epoch forgetting of +31.8% reflects a single seed that overshot (val MSE bottoms at epoch 7 before diverging); its best-epoch MSE is 0.132, essentially unchanged from ZS, qualitatively consistent with the N=5k trajectory analysis (Fig. 1, right panel). ΔR^2 and forgetting are therefore not co-determined by the same encoder layers. The top-3 of 6 layers are sufficient to restructure the encoder toward the trained objective (producing ΔR^2>0), but the lower 3 layers are required for the encoder to recover pre-trained task quality. The additional ΔR^2 gain from N=3 to N=6 (+0.48 → +0.63) is modest and has much higher variance (0.115 → 0.378), consistent with the lower layers contributing primarily to task recovery rather than to the task-aligned decodability shift.

This finding sharpens the central claim: the frozen-encoder control (Appendix M) rules out data-distribution confound; the N=3 result rules out a whole-encoder-restructuring account.

Layer-wise drift pattern. The N=3 versus N=6 contrast reveals a within-encoder dissociation: the *top 3 layers* suffice to reorient encoder geometry toward the trained objective (ΔR^2>0, 10/10 seeds, CKA≈0.66), but the *lower 3 layers* are required for task-performance recovery (N=3: forg. −6.5%, 9/10 negative; N=6: forg. −6.7%). Both N=3 and N=6 operate in the same deeply-negative-R^2 regime; the ΔR^2>0 signal at N=3 is a geometric reorientation, not a decodability gain, consistent with the main-text framing of frozen-probe ΔR^2 as directional evidence, not absolute evidence. (Probe-design robustness of Appendix M applies to both.)

Per-layer CKA breakdown. We computed per-layer CKA for condition B (N=6 full unfreeze, n=10k, h=96, 10 CUDA seeds). The bottom 3 layers are expected to drift more (lower CKA) than the top 3 under either the specialization or gradient-flow account; the data cannot distinguish these explanations without gradient-norm measurements.

Layer	CKA (mean±std)	Group
1 (bottom)	0.441±0.118	bottom 3
2	0.473±0.109	bottom 3
3	0.514±0.097	bottom 3
4	0.581±0.088	top 3
5	0.647±0.076	top 3
6 (top)	0.718±0.064	top 3
Bottom 3 mean	0.476	
Top 3 mean	0.649	

Bottom layers drift more (mean CKA=0.476 vs. 0.649 for top 3), *despite* the top 3 being sufficient for ΔR^2>0 (N=3 analysis above, 10/10 seeds). The more parsimonious explanation is standard gradient-magnitude patterns: under full backpropagation, earlier layers receive larger weight updates, producing lower CKA mechanically. Functional specialization remains an alternative hypothesis requiring a gradient-norm-per-layer comparison to distinguish from the gradient-flow null. The global CKA=0.518 at N=6 (Table 14) is the aggregate of these per-layer values.

O Chronos-T5-Small on M4-Monthly: Full Diagnostic Results

This appendix provides the complete results for the cross-backbone validation on Chronos-T5-Small (65M parameters, T5 seq2seq architecture with tokenised cross-entropy loss) fine-tuned on M4-Monthly (200 series, lookback=96, horizon=18). The gate-pass is 84.5% improvement over Linear baseline.

Protocol. 10 CUDA-deterministic seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), batch size 32, lr=1e-5, AdamW with weight decay 0.01, gradient clipping at 1.0, 20 epochs with early stopping (patience 5). Evaluation: batched median prediction (20 samples) on 200 held-out test windows.

Condition B (full fine-tune), n=500 **(10 seeds).**

	Forgetting%	CKA	ΔR^2	R^2(PT)
Mean±std	+1.6±3.9	0.999±0.001	−0.001±0.002	+0.068±0.025
Seeds neg/pos	4/10 neg forg	—	3/10 pos	**10/10 pos**

At n=500 the encoder is essentially unchanged (CKA≈1). The positive R^2(PT) confirms that Chronos's pre-trained T5 encoder representations carry decodable forecasting structure, providing a second positive-floor anchor distinct from Moirai.

Condition B (full fine-tune), n=10**k (10 seeds).**

	Forgetting%	CKA (ES)	ΔR^2	R^2(PT)
Mean±std	+0.5±2.7	0.983±0.021	≈0	+0.047±0.016
Seeds positive	6/10 forg	—	1/10	**10/10 pos**

At n=10k, early-stopped CKA averages 0.983 (vs. 0.999 at n=500). Only 4/10 seeds improve over zero-shot (best epoch>0, those 4 show CKA≈0.95–0.97); the remaining 6/10 revert to pre-trained weights under early stopping (best_epoch=0, CKA=1.000 by definition). The strong gate-pass (84.5%) means Chronos's encoder is already near-optimal for M4-Monthly, leaving minimal room for fine-tuning improvement. While 6/10 seeds show positive (worse) early-stopped forgetting, the magnitude (+0.5±2.7%) is within the noise floor established by the frozen-encoder control (condition D: +0.6±5.3%); a sign test is not statistically distinguishable from chance (p=0.75, two-sided binomial), confirming the 6/10 pattern reflects seed-level variation rather than systematic degradation.

Final-epoch comparison (protocol robustness). Training runs to epoch 20 regardless of early stopping; we report CKA at the final training epoch to address whether encoder stability is intrinsic or an artefact of early-stopping selection:

	Final-epoch CKA	Final-epoch Forg.%	n
n=500 (10 seeds)	0.968±0.009	+3.1±5.8	500
n=10k (10 seeds)	0.955±0.006	+3.4±5.2	10k

All 10 seeds at n=10k show final-epoch CKA>0.94 (min=0.945, max=0.964), confirming that encoder stability is *intrinsic* rather than an artefact of early-stopping selection. The mild CKA decline from 0.999 (early-stopped) to 0.955 (final-epoch) represents continued gradient pressure over 20 epochs, but remains far above the restructuring threshold observed on Moirai (CKA=0.52 on Small, 0.25 on Base). Final-epoch forgetting is +3.4±5.2%, mild degradation vs. the near-zero early-stopped result, but qualitatively different from Moirai-Small's +14% at n=500.

Mild CKA decline interpretation. The 0.999→0.955 CKA trajectory (early-stopped→final-epoch at n=10k) confirms a small but real gradient signal from the fine-tuning objective. We do *not* interpret this as "no drift"; rather, the drift is quantitatively minor (CKA>0.94) and functionally benign (final-epoch forgetting +3.4% vs. Moirai-Small's +14% at n=500). The value-gate is consistent with this: when pre-trained features are already near-optimal (84.5% advantage), the fine-tuning gradient has little room to improve and limited capacity to harm.

Condition D (frozen encoder), n=10k (10 seeds). Forgetting: $+0.6$±5.3% (5/10 negative). CKA=1.000 (encoder frozen by definition), ΔR^2=0. Decoder-only adaptation is unreliable: neither consistently better nor worse than zero-shot across seeds.

Random-init control (6 seeds). CKA=0.000 (expected: random weights have no similarity to pre-trained). R^2(PT)<0 (-0.004±0.005), confirming that positive-floor R^2 on the pre-trained encoder requires the learned representations, not the architecture alone. Training loss remains high (>8.0) and predictions are incoherent, confirming that Chronos's pre-trained representations are load-bearing.

Comparison with Moirai. The contrast validates the value-gate as a regime indicator: *Moderate gate-pass* (Moirai/ETTh2, 28%): encoder restructures aggressively (CKA=0.52), probe asymmetry emerges (ΔR^2=+0.67, orthogonal<0), forgetting is capacity- and protocol-dependent. *Strong gate-pass* (Chronos/M4, 84.5%): encoder is stable (CKA≥0.95 even at final epoch), no measurable probe shift, forgetting near-zero under early stopping and mild (+3.4%) at final epoch; fine-tuning is unnecessary but not harmful. Both outcomes are correctly predicted by the gate criterion, demonstrating that the diagnostic toolkit generalises across architecturally distinct backbones. The probe asymmetry mechanism (task-specific ΔR^2>0) requires encoder restructuring to produce a measurable signal; on Chronos, encoder stability means the asymmetry is not observable; this is a correct null prediction, not a failure of the diagnostic.

O.1 ILI Gate Check

Moirai-Small zero-shot on ILI (national illness, 966 weekly observations, 7 features, h=24, lookback=104) yields gate-pass=57% (ZS MSE=1.183, Linear MSE=2.748, per-feature normalisation by context statistics).

Full probe results (10 CUDA seeds, conditions B/D). Table 15 reports per-seed aggregate diagnostics. All 10 seeds show ΔR^2>0 (aggregate: $+0.123$±0.046; Wilson CI $[0.72, 1.00]$) and forgetting <0 (-80.7±0.9%; 10/10 negative, meaning fine-tuning dramatically improves on ZS). CKA=0.478±0.022 places ILI in the restructuring regime, not the predicted intermediate regime (0.90–0.95). Condition D (frozen encoder): CKA=1.000, ΔR^2=0, forgetting=0% in all 10 seeds, confirming encoder weight updates drive all changes.

Table 15: ILI per-seed results (condition B, Moirai-Small, h=24, lookback=104). Aggregate ΔR^2 is the mean across 7 univariate features.

Seed	CKA	ΔR^2	Pos. features	Forg.%	Drift
42	0.465	+0.211	6/7	−81.5	5.61
101	0.465	+0.123	3/7	−80.9	5.34
123	0.511	+0.087	3/7	−82.3	5.13
202	0.461	+0.169	5/7	−79.6	5.67
303	0.498	+0.169	7/7	−80.9	5.49
456	0.454	+0.066	5/7	−80.2	5.63
777	0.498	+0.074	4/7	−80.1	5.36
789	0.460	+0.144	6/7	−79.8	5.49
888	0.459	+0.111	4/7	−79.7	5.24
999	0.512	+0.080	4/7	−82.0	5.25
Mean±std	0.478±.022	+0.123±.046	4.7/7	−80.7±0.9	5.42±.21

Per-feature breakdown. OT (target feature) shows the largest ΔR^2 ($+0.73$±0.16, 10/10 positive); AGE 0–4 also shows consistent gain ($+0.15$±0.05, 10/10). AGE 5–24 shows slight negative signal (-0.07±0.09, 3/10 positive), consistent with the task-specificity interpretation: features less aligned with the forecasting objective show reduced or negative decodability shift.

Interpretation. The pre-registered prediction (CKA 0.90–0.95, intermediate behaviour) is partially falsified: CKA is in the restructuring regime (0.478), not intermediate. The probe-asymmetry prediction ($\geq$7/10 seeds $\Delta R^2 > 0$) is confirmed (10/10). This revises the gate-pass spectrum: the restructuring-to-stability transition occurs above $\approx$80% gate-pass, not at 50–70%. ILI's low CKA reflects the small dataset (966 timesteps, $\approx$450 training windows per feature) which permits aggressive restructuring within 20 epochs.

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